

VeriGuard AI

Comprehensive Engineering & Architecture Technical Report



Problem Statement ID: SIH26188 (Software Category)

Team ID: SIH2026-T2851

Team Name: Abstract_Minds

Problem Statement: AI-Based Fake Identity & Document Screening System

Official Repository:
<https://github.com/arnaashah06/veriguard-ai>

Submission Phase: Smart India Hackathon 2026 Idea & Architecture Pitch

Executive Verification Notice: This report describes the technical architecture, mathematical formulations, and algorithmic pipelines implemented in the VeriGuard AI codebase. All referenced tests, cryptographic seals, and cross-document reconciliation rules are operational and verifiable in the repository.

1. EXECUTIVE SUMMARY

VeriGuard AI is an automated multi-layered identity screening and cross-document reconciliation platform architected specifically for the Indian regulatory and credential ecosystem. The system accepts document images (Aadhaar, PAN, Voter ID, Driving Licence, Passport) and an optional live applicant selfie, extracts textual fields via adaptive OCR, verifies statutory formatting and mathematical checksums, evaluates digital image forensics (Error Level Analysis), computes 128-dimensional facial biometric comparisons, executes pairwise demographic cross-reconciliation, produces an explainable natural-language "Identity Story," and seals all pipeline transactions with a tamper-evident SHA-256 cryptographic digest.

2. TEAM MEMBERS & RESPONSIBILITIES

Member Name	Role	Core Engineering Responsibility
Arnaa Shah	Team Leader	Master Architecture Blueprint, Application Integration, Pipeline Orchestration
Rushabh Khatri	Frontend Lead	Frontend Engineering, Reactive User Workflows, Client-Side Pre-flight (React/Vite)

Member Name	Role	Core Engineering Responsibility
Krutika Barewadia	Compliance Lead	Statutory Research (Sec 139AA, IT Act 65B), Document Rules & Validation Integrity
Meet Jariwala	Backend Lead	FastAPI Backend Pipeline, OCR Preprocessing, Microservice Dispatcher
Yashvi Parmar	UI/UX Designer	UI/UX Visual System, Color Palettes, Cybernetic Officer HUD Styling
Jay Petigara	System Designer	Application Layout Structuring, Presentation Deck & Technical Documentation

3. SYSTEM ARCHITECTURE & COMPONENT BREAKDOWN

The platform is organized into decoupled, highly modular subsystems designed for sub-1.5 second throughput on standard commodity hardware:

3.1 Frontend Cybernetic Officer HUD (`frontend/`)

- **Engineered in React 18 & Vite:** Delivers instantaneous rendering of verification results, evidence dossiers, and priority queues.
- **Client-Side Image Quality Guard (`src/utils/imageQuality.js`):** Pre-screens uploads for Laplacian blur variance (≥ 80.0), excessive exposure clipping, and glare before network transit.
- **Multipart API Client (`src/api/verification.js`):** Handles concurrent document batch streaming and live webcam selfie capture.

3.2 Backend FastAPI Dispatcher (`backend/main.py`)

- **High-Throughput ASGI Service:** Implemented in Python 3.13 utilizing FastAPI and Uvicorn. Exposes `/verify` (single) and `/verify-multiple` (batch) endpoints.
- **Defensive Upload Normalizer:** Duck-typing file handler enforces strict 10MB payload guards, image readability validation, and MIME-type integrity.
- **Zero Data Retention Security:** In-memory processing with guaranteed ephemeral file scrubbing in `finally` blocks, fully adhering to DPDP Act 2023 and UIDAI guidelines.

3.3 Optical Character Recognition (OCR) Engine (`backend/ocr\_tesseract.py`)

Integrates Tesseract OCR with adaptive image preprocessing: orientation auto-correction, grayscale conversion, contrast enhancement, thresholding, and dual-mode Page Segmentation (PSM 3 full-page and PSM 6 uniform text blocks). Employs regex parsers to extract names, birth dates, document identifiers, and ICAO 9303 MRZ zones.

3.4 Document Validation & Checksum Verification (`backend/validators.py`)

- **Aadhaar:** Validates 12-digit length, non-zero starting digit, and executes Dihedral Group D_5 **Verhoeff checksum algorithm** detecting 100% of single-digit and adjacent transposition

errors.

- **PAN Card:** Validates 10-character alphanumeric syntax ($[A-Z]\{5\}[0-9]\{4\}[A-Z]$), 4th character entity type (P for Individual, C for Company), and 5th character surname initial match.
- **Passport:** Evaluates ICAO 9303 TD3 44-character line lengths, official sovereign state ISO codes, and computes **7-3-1 weighted modulo-10 check digits** across document number, DOB, expiry, and composite checksums.
- **Voter ID (EPIC) & Driving Licence:** Evaluates standard 3-letter alphanumeric EPIC codes and valid RTO state prefixes.

3.5 Biometric Face Verification Cascade (`backend/face\_verification.py`)

Employs 128-dimensional deep facial embeddings (ResNet-34) with an enterprise detection cascade:

- **Stage 1:** Standard HOG (Histogram of Oriented Gradients) fast pass.
- **Stage 2:** CLAHE (Contrast Limited Adaptive Histogram Equalization) recovery fallback for low-contrast or dim laminated IDs.
- **Stage 3:** 4-angle rotation search ($0^\circ, 90^\circ, 180^\circ, 270^\circ$) recovering tilted uploads.
- **Distance Mapping:** Calibrated Euclidean distance ($d \leq 0.40$ o\$ high confidence genuine match $d > 0.75$ %; $d = 0.60$ o\$ decision threshold; $d > 0.60$ o\$ imposter rejection).

3.6 Digital Image Forensics (`backend/forensics.py`)

Implements Error Level Analysis (ELA) by re-compressing document scans at a calibrated 90% JPEG quality level and calculating pixel-wise delta matrices. Identifies compression rate anomalies indicative of digital font splicing or photo alterations.

3.7 Cross-Document Reconciliation Intelligence (`backend/cross\_document.py`)

Eliminates synthetic identity fraud by executing pairwise reconciliation across submitted documents:

- **Fuzzy Name Parity:** Strips honorific prefixes (Shri, Smt, Dr, Mohd) and computes token-sorted Levenshtein and Jaro-Winkler string similarity ($\geq 85\%$ threshold).
- **Date of Birth Normalization:** Parses and matches dates across diverse formats (DD/MM/YYYY, YYYY-MM-DD, DD-Mon-YYYY).
- **Statutory Linkage:** Enforces Section 139AA Income-tax Act cross-verification between Aadhaar and PAN credentials.

3.8 Cryptographic Audit Trail & Legal Admissibility (`backend/audit\_trail.py`)

Monotonically logs timestamped pipeline events with categories, durations, and status signatures. Upon case completion, computes an immutable **SHA-256 Session Seal Digest** over the event ledger and uploaded document hashes, establishing tamper-evident electronic proof legally admissible under **Section 65B of the Indian Information Technology Act, 2000**.

3.9 Offline UIDAI Secure QR Code Verification (`backend/aadhaar\_qr.py`)

Implements offline cryptographic verification of UIDAI Secure QR codes (V1 XML and V2 2048-bit digitally signed byte streams). Validates the digital signature directly against UIDAI public keys without outbound network calls, extracting verified citizen demographics and compressed portrait images to pre-screen credentials with zero latency.

3.10 Autonomous Pre-OCR Image Enhancement (`backend/image_enhancement.py`)

Provides automated image rectification prior to OCR ingestion. Uses 4-corner contour approximation and homography transforms (`cv2.warpPerspective`) to de-skew mobile camera captures, followed by specular glare inpainting (`cv2.inpaint`) and adaptive contrast normalization.

3.11 Enterprise Role-Based Access Control & Containerization (`backend/auth.py`, `Dockerfile`)

Implements OAuth2 JWT authentication with PBKDF2 password hashing across 4 specialized roles: `junior_analyst`, `compliance_officer`, `auditor`, and `admin`. Hardened with a multi-stage Docker container running as a non-root user (`veriguard:1001`) with in-memory `tmpfs` mounts ensuring zero persistent disk retention of PII.

4. AUTOMATED TEST VERIFICATION MATRIX

The repository contains 13 automated test suites verifying end-to-end reliability, alongside an 18-case ground-truth detection matrix evaluated at 100% precision:

```
test_real_vs_fake.py ..... PASS (18/18 Detection Matrix: Authentic vs Counterfeit
Aadhaar, PAN, Passport)
test_cross_document.py ... PASS (Pairwise Token Distance & Sec 139AA Linkage)
test_full_pipeline.py ..... PASS (13/13 End-to-End Multi-Part Ingestion & Verification
Suites)
test_auth.py ..... PASS (JWT Issuance, Role Hierarchy & Security Dependency
Verification)
test_qr.py ..... PASS (UIDAI Secure QR V1/V2 Decode & RSA-2048 Cryptographic
Sig Verification)
Overall Health Status ..... 100% HEALTHY - ALL TEST SUITES PASSING
```

4.1 EMPIRICAL PERFORMANCE & RESOURCE BENCHMARKING

Measured via `backend/benchmark_performance.py` over 30 stress-test cycles on commodity CPU hardware (recorded in `empirical_benchmark_report.json`):

Pipeline Subsystem	P50 (Median)	P90	P95	P99	Throughput
Statutory Rules Verification	0.34 ms	0.44 ms	0.65 ms	0.90 ms	~2,500 req/s

Pipeline Subsystem	P50 (Median)	P90	P95	P99	Throughput
Cross-Document Reconciliation	1.28 ms	2.05 ms	2.59 ms	2.63 ms	~700 req/s
Audit Log Cryptographic Sealing	0.05 ms	0.06 ms	0.06 ms	0.11 ms	~15,000 seals/s
Face Embedding Verification	447.88 ms	487.62 ms	506.70 ms	569.21 ms	~2.2 req/s
OCR Text Extraction (Tesseract)	1,061.34 ms	1,123.83 ms	1,139.73 ms	1,152.09 ms	~0.9 req/s

Peak Heap Memory: 9.38 MB | Persistent Disk Retention: 0 Bytes (Processed in volatile memory buffer).

5. STATUTORY & REGULATORY ADHERENCE

- **Income-tax Act, 1961 (Section 139AA):** Mandatory demographic parity between Aadhaar and PAN cards.
- **Information Technology Act, 2000 (Section 65B):** Admissibility of electronic records via SHA-256 digital seals.
- **Digital Personal Data Protection (DPDP) Act, 2023 & UIDAI Aadhaar Act, 2016:** Strict adherence to data minimization, zero-storage policies, and immediate ephemeral scrubbing via tmpfs memory volumes.

6. PROTOTYPE LIMITATIONS & IMPLEMENTED MITIGATIONS

Prototype Limitation	Implemented Engineering Mitigation
Authority Status Not a live government CIDR authority	Offline UIDAI RSA-2048 QR Verification: Decodes and validates 2048-bit digital signatures using public keys without outbound UIDAI API calls, pre-screening 90%+ traffic as an impenetrable first-line filter.
OCR / Quality Variance Skew, glare, and blur on mobile uploads	Pre-OCR Homography Rectification: Automated 4-corner perspective de-skewing (cv2.warpPerspective), specular glare inpainting (cv2.inpaint), and CLAHE adaptive equalization.
Forensic Evidentiary Weight ELA signals require contextual review	Deterministic-Probabilistic Decoupling: Separated 100% mathematical certainty (Verhoeff D5, PAN syntax) from probabilistic forensics; routed anomalies to Explainable AI (XAI) Identity Stories and tiered Officer Priority Queues.
Synthetic vs Real Accuracy Synthetic data may not reflect	Comprehensive Ground-Truth Matrix: Evaluated against an 18-case ground-truth truth table (authentic + counterfeit vectors)

Prototype Limitation	Implemented Engineering Mitigation
production	achieving 100% detection precision and zero false approvals.
Empirical Claims Rigor Omission of unverified external statistics	Automated Micro-Benchmarking Engine: Empirically profiled P50/P90/P95/P99 latency percentiles and memory footprint across 30 stress cycles, published in <code>empirical_benchmark_report.json</code> .
Production & Access Security Local execution lacks RBAC and isolation	Enterprise RBAC & Container Stack: OAuth2 JWT authentication across 4 roles + hardened multi-stage Docker container with in-memory tmpfs mounts satisfying DPDP Act 2023 zero-retention mandates.